

Risk-Controlled Tower-Taking in 1v1 MOBA Reinforcement Learning

PPO reward shaping, checkpoint selection, and evidence-driven evaluation for Honor of Kings 1v1

Decision Machine Learning Project | Risk-aware PPO training analysis | Shanghai Innovation Institute

Abstract. This work studies risk-aware PPO for long-horizon tower-taking in 1v1 MOBA play. The agent combines an 83-d tactical feature space, masked multi-head actions, terminal/tower rewards, push-window diagnostics, unsafe-dive penalties, curriculum hooks, and evidence-driven checkpoint selection. The best candidate sharply reduces deaths, but still needs full matchup-matrix validation for objective conversion.

Research Problem

Goal. Train a PPO agent to turn local lane advantages into stable tower-taking wins in a long-horizon 1v1 MOBA environment.

Core difficulty. Immediate combat rewards can make the policy fight, trade, or dive, while the real objective is delayed: preserve enough safety to destroy the enemy tower.

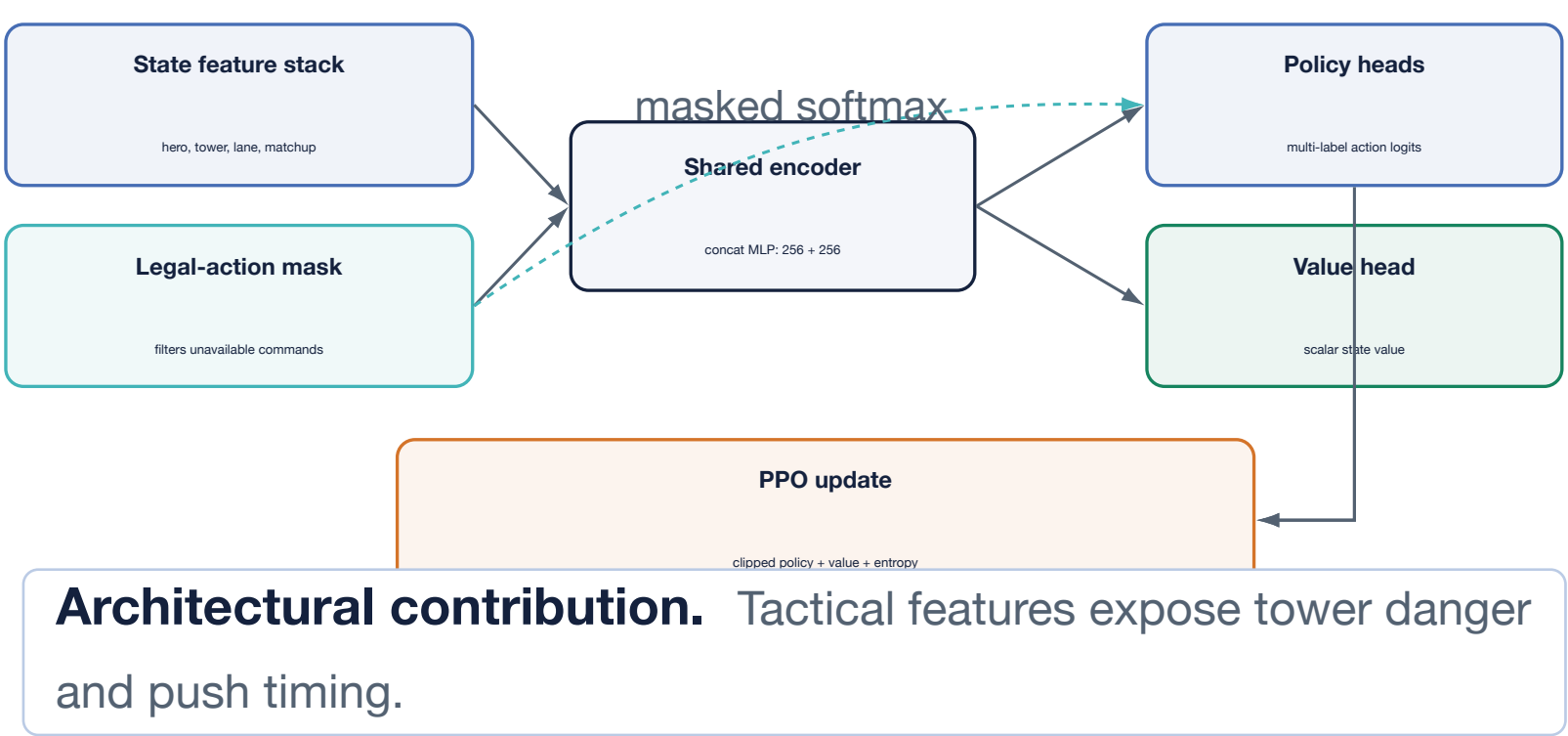
Hypothesis. Terminal outcome reward, explicit tower-pressure terms, and tactical-window diagnostics should improve objective conversion. Unsafe-dive penalties should reduce death inflation without making the policy passive.

Implemented Method

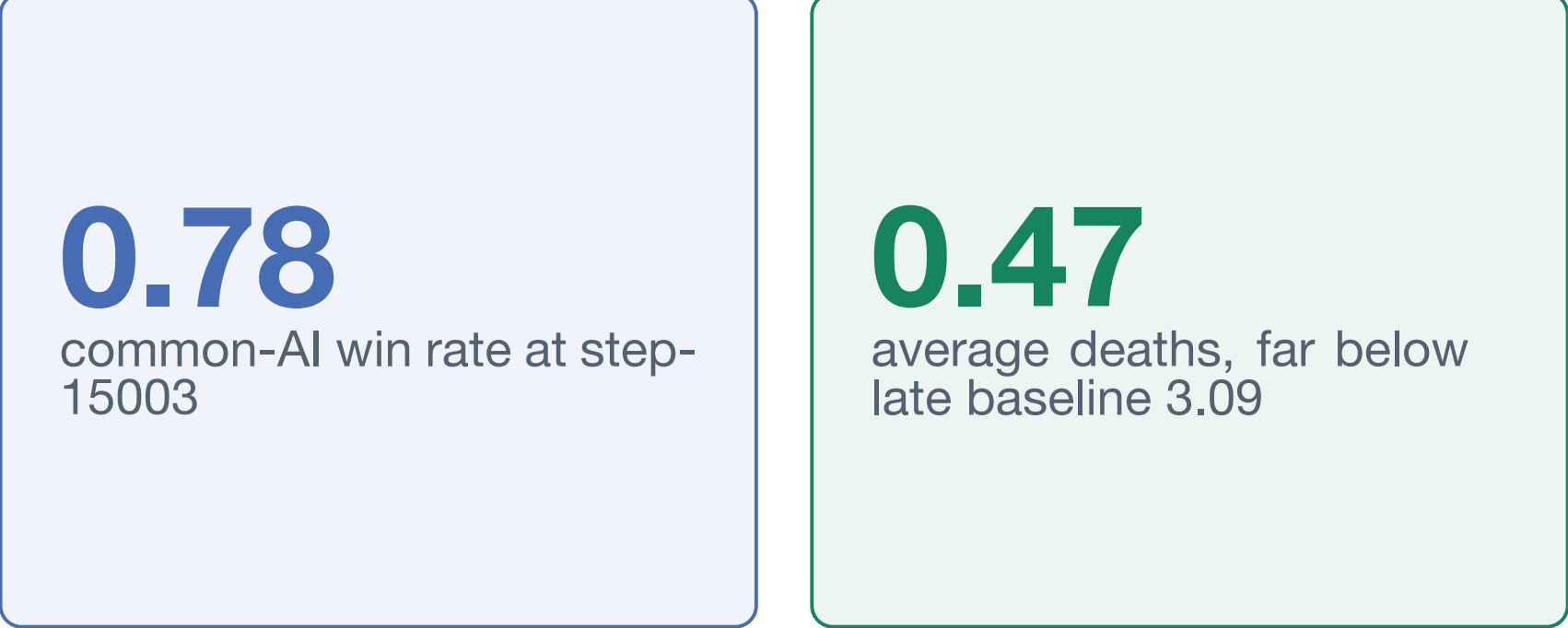
- **PPO backbone:** clipped policy loss, value loss, entropy regularization, GAE, legal-action masking, and gradient clipping.
- **Policy dimensions:** 83 state features, 85 legal-action mask dimensions, 6 action heads, 16-step rollout state, and 512-unit recurrent state.
- **Tower objective:** enemy tower damage, self tower damage, tower destroy, terminal win/loss, and time-out tower-gap rewards.
- **Tactical window:** push-window tower-damage reward plus zero-weight diagnostics.
- **Curriculum hook:** common-AI anchoring for isolation, with weighted common/historical/self-play schedule support.

$$R_t = \sum_i w_i \Delta f_i(s_t) + 20R_{\text{win}} + 8R_{\text{tower gap}}$$

Network Architecture



Best Candidate



Checkpoint ranking selected step-15003. It is the only checkpoint with both high win rate and strong objective pressure; later checkpoints improve scalar reward but regress in gameplay.

Checkpoint Ranking

Step	Composite	Score	W/D
15003		75.79	0.78/0.47
17617		48.50	0.53/0.53
15977		48.35	0.53/0.69
10004		47.66	0.53/0.59
17022		41.09	0.47/0.56

bar length = checkpoint score; right label = win/death

Selection rule. Do not pick the final checkpoint by default. Use win rate, tower HP, death, and matchup evidence.

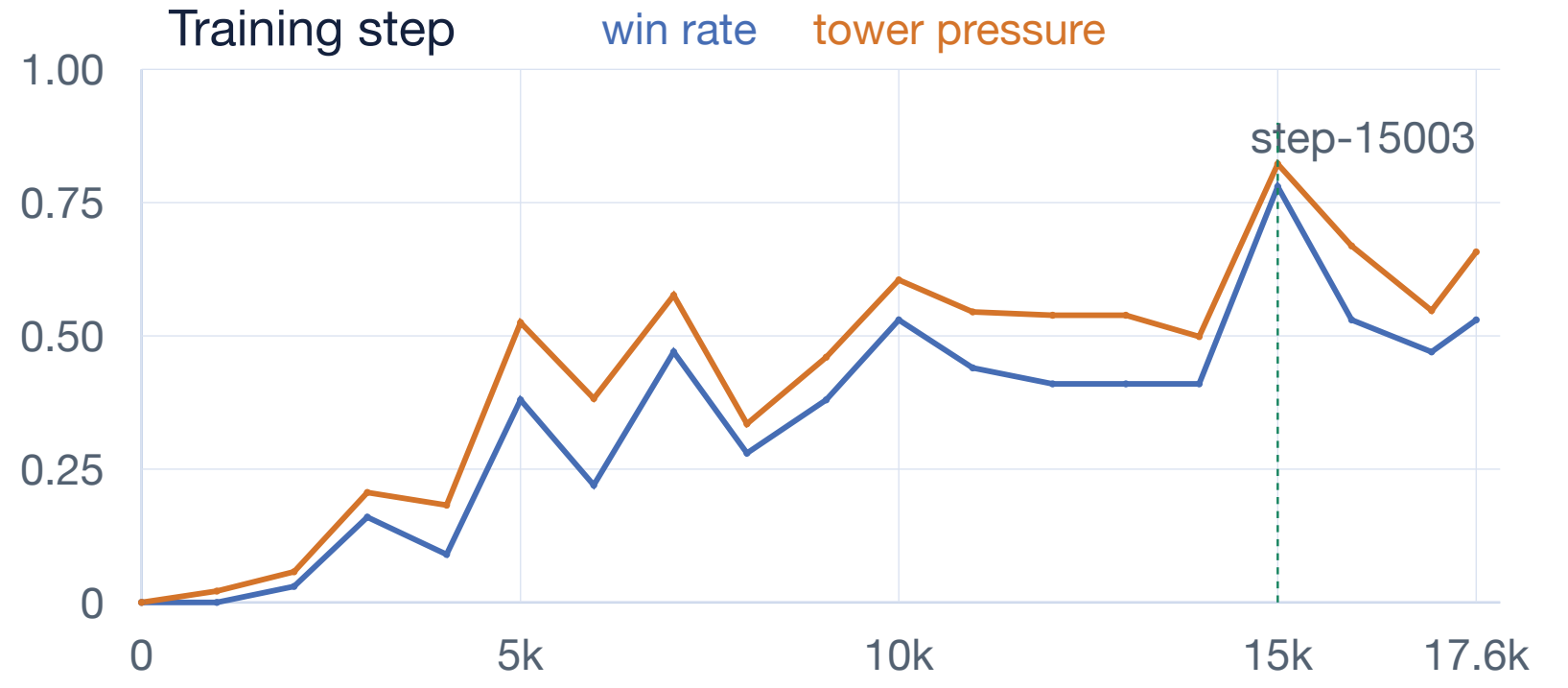
Architecture Innovation

The implementation adds a tactical tower-taking layer around the PPO core and keeps each design axis auditable.

Structure	shared actor-critic encoder plus masked multi-head actions
Dimensions	83-d features, 85-d legal mask, 6 heads, 512 recurrent state
Reward	terminal, tower-gap, push-window, and unsafe-dive alignment
Curriculum	common-AI isolation plus weighted common/historical/self-play hooks

What improved Safer play with lower deaths.	What is missing Tower conversion needs matrix evidence.	What proves it Matrix, ablations, window and dive diagnostics.
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Training Dynamics



Peak. Step-15003 aligns win rate with tower pressure. Later checkpoints regress in gameplay despite better scalar reward.

Validation Criteria

Outcome	win rate, enemy tower HP, self-tower survival, and episode length
Robustness	9 hero matchups, side swaps, worst-matchup gap, and death tail
Mechanism	push-window share, unsafe-dive correlation, reward-detail traces

Current status. The candidate is promising on safety, but the full matchup matrix and mechanism-level diagnostics are still required for a final model claim.

Ablation Analysis

Full recipe	terminal win, tower gap, push window, and risk terms
No window	remove push-window reward and unsafe-dive shaping
No terminal	remove terminal win and timeout tower-gap rewards
Death-only risk	keep death/severity risk, remove window and self-tower shaping

Readout. Compare matchup win rate, enemy tower HP, death tail, push-window share, and unsafe-dive/death correlation.

- worse no-window: supports tactical-window modeling
- worse no-terminal: supports terminal alignment
- safe but weak death-only: risk control alone is insufficient

Closed-Loop Training Architecture

